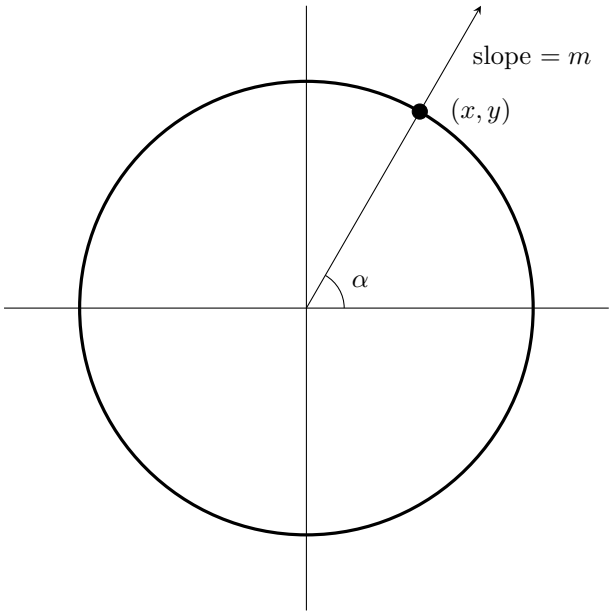
MATH 195: 10/22
CIRCLES AND TRIGONOMETRY

Sine, cosine, and tangent.

An angle determines a unique point on the *unit circle*. We can think of the coordinates of a point on the circle as a function of the angle. The *cosine* function outputs the x-coordinate and the *sine* function outputs the y-coordinate. The angle also determines the slope of the line goig out, which we call the *tangent* of the angle.



$$\cos\alpha = x$$
$$\sin\alpha = y$$
$$\tan\alpha = m = \frac{y}{x} = \frac{\sin\alpha}{\cos\alpha}$$

Special angles.

In general, it is difficult to write a formula for the output of a trig function. But we can use geometry to work out the values of some special angles.

α	cos	sin	tan
0			
$\pi/6$			
$\pi/4$			
$\pi/3$			
$\pi/2$			

Other quadrants.

The sign—positive or negative—or a trig function is determined by its quadrant. The magnitude is determined by the value of the function at its reference angle.

Other trig functions.

There are other trig functions which you can define based on sine and cosine. The three you still sometimes see today are the *secant*, *cosecant*, and *cotangent*.

$$\sec\alpha = \frac{1}{\cos\alpha}$$
$$\csc\alpha = \frac{1}{\sin\alpha}$$
$$\cot\alpha = \frac{1}{\tan\alpha} = \frac{\cos\alpha}{\sin\alpha}$$

Practice problems

(1) An angle α in quadrant I has a tangent of $6/7$. Determine its sine and cosine.
(2) An angle α in quadrant II has a cosine of $-3/5$. Determine its sine and tangent.
(3) A genie tells you that
$$\sin(\pi/12) = \frac{\sqrt{6}-\sqrt{2}}{4}.$$
Determine the values of
$$\sin(11\pi/12) \quad \sin(13\pi/12) \quad \sin(23\pi/12)$$
(4) Also determine $\cos(\pi/12)$ and $\tan(\pi/12)$.
(5) Furthermore determine the values of
$$\begin{array}{lll}\cos(11\pi/12) & \cos(13\pi/12) & \cos(23\pi/12)\\ \tan(11\pi/12) & \tan(13\pi/12) & \tan(23\pi/12)\end{array}$$
(6) Find all angles where a trig function has an output of 1. Which trig functions are 1 there?
(7) Find all angles where a trig function has an output of 0. Which trig functions are 0 there?
(8) What are the domain and range of sin and cos?
(9) What are the domain and range of tan?
(10) For which angles α is $\cot\alpha$ undefined?
(11) Explain why $|\sec\alpha|$ and $|\csc\alpha|$ are always ≥ 1.
(12) Your friend claims that $\cos(-\alpha) = \cos(\alpha)$ for any angle α. Either find a counterexample or explain why they are correct.
(13) Your friend claims that $\sin(-\alpha) = -\sin(\alpha)$ for any angle α. Either find a counterexample or explain why they are correct.
(14) Determine a formula for what $\tan(-\alpha)$ is.
(15) Explain why $\sin(2\pi+\alpha) = \sin(\alpha)$ and $\cos(2\pi+\alpha) = \cos(\alpha)$ for any angle α.
(16) Explain why $\tan(\pi+\alpha) = \tan(\alpha)$ for any angle α.